

J-Anne Yow

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Summary

CTO at Phridom AI, leading embodied AI and robotics development across robot data infrastructure, VLA training and evaluation, and system architecture. PhD-trained in robotics and AI at Nanyang Technological University, with experience spanning robot learning, multimodal perception, and real-world robotic systems.

Work Experience

Chief Technology Officer, Phridom AI – Shenzhen, China Dec 2025 – present

- Lead Phridom AI's data strategy for embodied AI, spanning multimodal data acquisition, alignment, validation, dataset analysis, and using model performance and data gaps to guide subsequent data collection.
- Drive the company's VLA model stack, including training and evaluation pipelines, observation/action representation design, model benchmarking, and adaptation of emerging embodied AI models to real robot systems.
- Define the technical product direction for Phridom AI's robotics platform, translating data collection, training, evaluation, and deployment workflows into scalable platform capabilities.
- Define the robotics research and engineering roadmap with the co-CTO and lead a team of 10+ engineers, owning key architectural decisions, technical prioritization, hiring, and team development.

Research Assistant, Nanyang Technological University – Singapore Aug 2021 – Nov 2025

- Managed ROS/ROS2-based system integration and hardware-software interfacing, collaborating with research staff and students to define technical goals and evaluate prototypes in simulation and on physical robots.
- Managed project planning, resource coordination, and ethical compliance, removing operational bottlenecks such as compute limitations to keep development on schedule.
- Mentored and scoped eight undergraduate projects, defining research objectives, guiding technical execution and skill development.
- Drafted and contributed to competitive funding proposals to support ongoing research and new initiatives.

Education

Nanyang Technological University, Singapore Sept 2021 – Mar 2026

Ph.D. in Mechanical Engineering

- **Supervisor:** Prof Ang Wei Tech
- **Dissertation:** Towards Personalized Robot Assistance: Integrating User Preferences in Robot-Assisted Feeding
- **Research Interests:** Human-Robot Interaction, Machine Learning, Foundation Models, Continual Learning

Nanyang Technological University, Singapore Aug 2017 – May 2021

BE in Mechanical Engineering with a Second Major in Business

- **GPA:** 4.92/5.00, First Class Honours
- **Awards:** Dean's List every Academic Year, Dr Leung Shiu Kee Gold Medal Award, ASEAN Undergraduate Scholarship

Research Experience

Robot-Assisted Feeding 2022 – 2025

- Led the technical delivery of an end-to-end adaptive feeding system, integrating perception, planning, and control into a safe, deployable platform for real-world elderly user trials.
- Developed goal-conditioned reinforcement learning policies in MuJoCo, integrating multimodal feedback (vision + force-torque sensing) for adaptive manipulation across variable food properties, achieving successful sim-to-real transfer.
- Built a language-driven adaptation pipeline, leveraging large language models to map user feedback into motion parameters for reliable, safe robot behaviour.

Grasping in Clutter 2020 – 2023

- Developed a point-and-click interface for human-in-the-loop robotic manipulation, integrating object segmentation, grasp ranking, and grasp cycling to streamline user input and improve selection efficiency in cluttered scenes.
- Developed a POMDP-based shared autonomy framework that queries users under intent uncertainty, reducing conflicting actions and enabling faster task completion.

Selected Publications

SAVR: Scooping Adaptation for Variable food properties via Reinforcement Learning <i>J-Anne Yow</i> , Wei Tech Ang IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)	Oct 2025
FRANC: Feeding Robot for Adaptive Needs and Personalized Care <i>J-Anne Yow</i> , Luke Toh, Yi Heng San, Wei Tech Ang IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)	Oct 2025
ORBiT: Optimizing Robot-Assisted Bite Transfer Leveraging a Real2Sim2Real Framework Sherwin Chan, <i>J-Anne Yow</i> , Yi Heng San, Vasanth Ravichandram, Yifan Wang, Lek Syn Lim, Wei Tech Ang IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)	Oct 2025
Shared Autonomy of a Robotic Manipulator for Grasping under Human Intent Uncertainty using POMDPs <i>J-Anne Yow</i> , Neha P Garg, Wei Tech Ang IEEE Transactions on Robotics (T-RO)	Nov 2023

Service

Reviewer: IEEE Robotics and Automation Letters (RAL), IEEE International Conference on Robotics and Automation (ICRA)

Skills

Programming: Python, C/C++

Robotics & Simulation: ROS/ROS2, MoveIt, MuJoCo, Isaac Lab

Machine Learning: Reinforcement Learning, Imitation Learning, PyTorch

Perception & Control: Vision, Force-Torque Sensing, Impedance/Admittance Control

Languages: English, Chinese, Malay